

Comprehensive Briefing: Computer Networks and IP Addressing Fundamentals

Summary

This document provides a **comprehensive, exam-oriented synthesis** of **Computer Networks (CN)** with a sharp focus on **IP addressing fundamentals**, aligned with the Unacademy GATE CSIT crash course perspective.

Computer Networks consistently contribute **~10 marks in GATE**, and IP Addressing alone guarantees **at least 2 marks**, often more when combined with subnetting and CIDR-based problems. The technical core of this briefing traces the **evolution of IPv4 addressing**—from early **fixed and classful models** to the modern **classless (CIDR) paradigm**—highlighting the **mathematical logic, design inefficiencies**, and **optimization principles** that every GATE aspirant must master.

Course Structure and Strategic Weightage

High-Priority Exam Topics (Five-Star Areas)

Topic	Typical Weightage	Key Sub-Topics
IP Addressing	≥ 2 Marks	Subnetting, Classless Addressing, CIDR
Error Control	Variable	CRC, Hamming Code
Flow Control	2 Marks (Every Year)	Stop-and-Wait, Go-Back-N, Selective Repeat
Transport Layer	~2 Marks	TCP Header, Wrap Around Time, Connection Phases
Medium Access	High	ALOHA, CSMA/CD
Switching	Variable	Packet Switching (formula-based)

Application Layer 1–2 Marks

DNS, HTTP, SMTP, FTP

Instructor Insight:

Completing the crash course thoroughly ensures **7–8 marks out of 10**, even before advanced problem-solving.

Mathematical Foundations of Networking



Bits, Bytes, and Powers of Two

- 1 Byte (B) = 8 bits (b)
- Address space calculations rely on powers of two:

Power	Size
2^{10}	1 KB
2^{20}	1 MB
2^{30}	1 GB
2^{40}	1 TB

N-Bit Range Rule

For any **n-bit binary number**:

- Range = **0 to $(2^n - 1)$**
- IPv4 octet (8 bits): **0–255**

Binary Weighting (8-bit)

Bit Position	Weight
b7 → b0	128 64 32 16 8 4 2 1

Quick GATE Shortcuts

- Continuous **1s from right**:
→ $1111_2 = 2^4 - 1 = 15$
- Continuous **1s from left**:
→ $11100000_2 = 128 + 64 + 32 = 224$

Address Space Partitioning Logic

If:

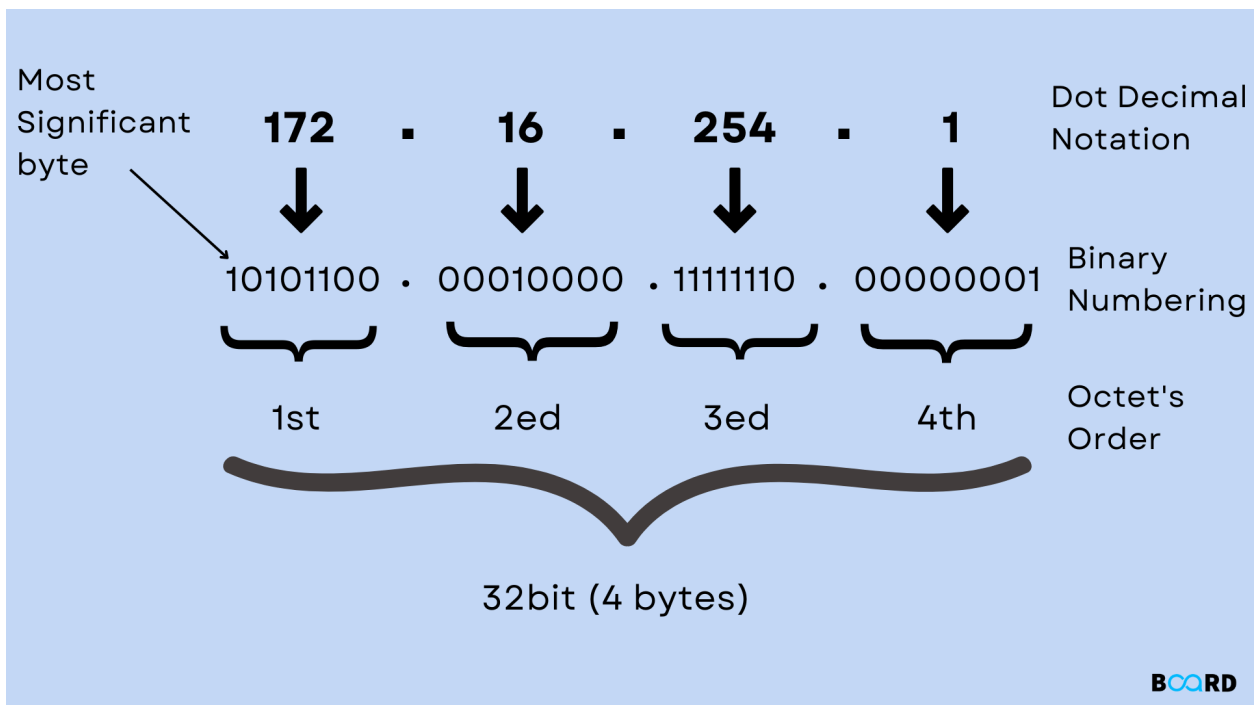
- Total bits = **n**
- Network prefix bits = **k**

Then:

1. Number of networks = 2^k
2. Hosts per network = 2^{n-k}
3. Total addresses = 2^n

This formula underpins **classful addressing, subnetting, and CIDR**.

Evolution of IP Addressing



Five Different Classes of IPv4 Addresses

Class	First Octet decimal (range)	First Octet binary (range)	IP range	Subnet Mask	Hosts per Network ID	# of networks
Class A	0 – 127	0XXXXXXXX	0.0.0.0-127.255.255.255	255.0.0.0	$2^{24}-2$	2^7
Class B	128 – 191	10XXXXXXXX	128.0.0.0-191.255.255.255	255.255.0.0	$2^{16}-2$	2^{14}
Class C	192 – 223	110XXXXXX	192.0.0.0-223.255.255.255	255.255.255.0	2^8-2	2^{21}
Class D (Multicast)	224 – 239	1110XXXX	224.0.0.0-239.255.255.255			
Class E (Experimental)	240 – 255	1111XXXX	240.0.0.0-255.255.255.255			





The 1980 Fixed Addressing Model

- IPv4 size: **32 bits**
- Structure:
 - Network ID: **8 bits**
 - Host ID: **24 bits**

Consequences

- Networks: $2^8 = 256$
- Hosts per network: $2^{24} \approx 16.7$ million
- Severe inefficiency for small organizations

Classful Addressing Paradigm

Classes	Network Bits	Host Bits	Networks	Hosts / Network	Intended Use
A	8	24	256	~16.7 M	Massive orgs (NASA)
B	16	16	~65 K	65,536	Mid-size (IBM, TCS)
C	24	8	~16.7 M	256	Small (Schools)

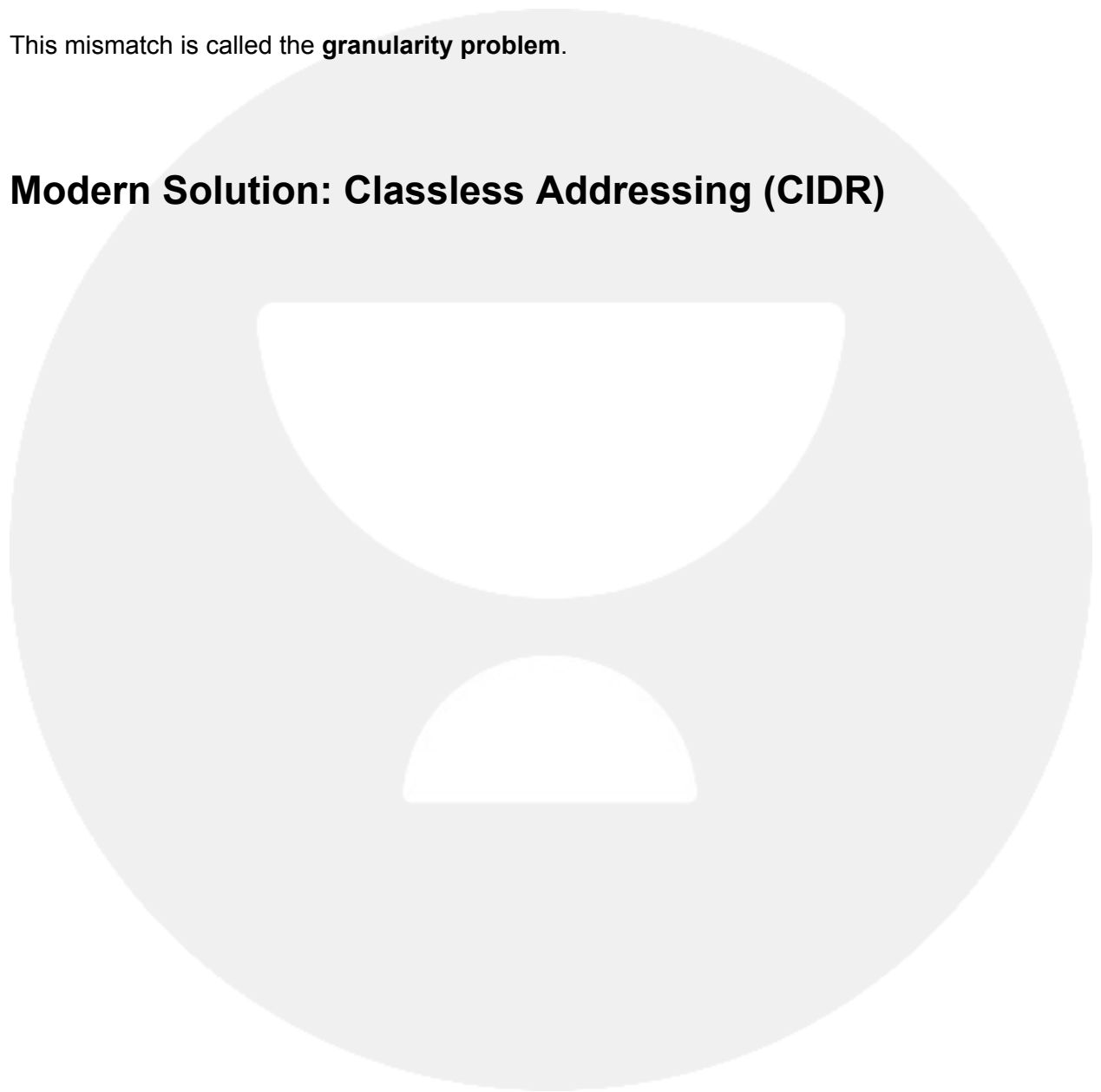
Why Classful Addressing Failed

- Organization needs **500 IPs**
→ Class C insufficient (256)

- Forced into Class B (65,536)
- **~65,000 IPs wasted**
- Organization needs **70,000 IPs**
 - Class B insufficient
 - Forced into Class A
 - **Millions wasted**

This mismatch is called the **granularity problem**.

Modern Solution: Classless Addressing (CIDR)



IPv4 CIDR Chart

$K = 1,024 \cdot M = 1,048,576$

IP Addresses	Bits	Prefix	Subnet Mask
1	0	/32	255.255.255.255
2	1	/31	255.255.255.254
4	2	/30	255.255.255.252
8	3	/29	255.255.255.248
16	4	/28	255.255.255.240
32	5	/27	255.255.255.224
64	6	/26	255.255.255.192
128	7	/25	255.255.255.128
256	8	/24	255.255.255.0
512	9	/23	255.255.254.0
1 K	10	/22	255.255.252.0
2 K	11	/21	255.255.248.0
4 K	12	/20	255.255.240.0
8 K	13	/19	255.255.224.0
16 K	14	/18	255.255.192.0
32 K	15	/17	255.255.128.0
64 K	16	/16	255.255.0.0
128 K	17	/15	255.254.0.0
256 K	18	/14	255.252.0.0
512 K	19	/13	255.248.0.0
1 M	20	/12	255.240.0.0
2 M	21	/11	255.224.0.0
4 M	22	/10	255.192.0.0
8 M	23	/9	255.128.0.0
16 M	24	/8	255.0.0.0
32 M	25	/7	254.0.0.0
64 M	26	/6	252.0.0.0
128 M	27	/5	248.0.0.0
256 M	28	/4	240.0.0.0
512 M	29	/3	224.0.0.0
1024 M	30	/2	192.0.0.0
2048 M	31	/1	128.0.0.0
4096 M	32	/0	0.0.0.0

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CIDR: 255.255.255.240/24

IP Range: 255.255.255.0
255.255.255.255

Total Hosts: 256

What Is CIDR

Class C Subnetting chart (CIDR & bitcounts)

CIDR	→	/25	/26	/27	/28	/29	/30
		↑	↑	↑	↑	↑	↑
Bitcounts/hosts	→	128	64	32	16	8	4
		↑	↑	↑	↑	↑	↑
# of networks	→	2	4	8	16	32	64

Useable hosts is Hosts - 2

Remember to allow 10% for growth

Write out chart 50 times

Made by Matt, Network+, Per Scholas

IPv4 BIT COUNTING SYSTEM

128 64 32 16 8 4 2 1

192.168.1.1

11000000.10101000.00000001.00000001

1.1.0.0.0.0.0.0

$128+64+0+0+0+0+0+0=192$

1.0.1.0.1.0.0.0

$128+0+32+0+8+0+0+0=168$

0.0.0.0.0.0.0.1

$0+0+0+0+0+0+0+1=1$

0.0.0.0.0.0.0.1

$0+0+0+0+0+0+0+0=1$

Cake Analogy

- **Classful**: Fixed cake sizes (2kg, 10kg, 20kg)
- **Classless (CIDR)**: Cake cut exactly as required

CIDR Fundamentals

- Entire IPv4 space treated as **one pool**:
 $2^{32} \approx 4.29$ billion addresses
- No predefined classes
- Blocks allocated as powers of two:
 - /25 → 128 IPs
 - /22 → 1024 IPs
- Allocation handled by **IANA → ISPs → Organizations**

Benefits

- Minimal IP wastage
- Flexible subnet sizes
- Scalable internet growth

Pedagogical Insights and Resources

Recommended Standard Textbooks

- **Data Communications and Networking** – Best all-round CN reference
- **Computer Networks** – Strong theoretical depth
- **Computer Networking: A Top-Down Approach** – Application-driven understanding

Instructor's Motivational Insight

“He who spends himself, the world searches for him on Google.”

This emphasizes that **deep conceptual mastery**, not superficial memorization, is what leads to **top-200 GATE ranks**.

